

COMP2850 Project Spec

Supermarket

A new online-only supermarket has requested that your team help them create their web application. They have recently expanded their warehousing provision and need a system which can help them manage stock and orders and get them out to their customers.

This has several sub-systems:

- The E-commerce side, where users create orders
- The warehousing side, producing picking lists and stock updates
- The marketing/management side, which highlights popular products and allows metrics to be produced to review performance of different areas.

These systems all need to work together to produce a single e-commerce solution.

E-Commerce

The supermarket is, at its core, an online shop. You will create a web app which serves the existing stock from the database to customers and allows them to create a shopping basket of products.

They should be able to check out, at which point a payment will be made and the delivery scheduled based on availability data from the warehouse.

Users should be able to see updates about their order, and any previous orders.

Warehousing

Orders will be passed to the relevant warehouse, where a picking list is produced. This will be accessed by workers on a tablet or mobile phone device who will pick the order (collect all the items in the order), noting any issues such as out-of-stock items or replacement items.

Once the picker has finalised the order, this updates for the user to show any substitutions and should also charge/refund differences in price caused by missing items or substitutions.

The stock database should be updated dynamically while workers are picking orders to ensure that stock is always correct as multiple people may be picking at once.

The warehousing team also processes incoming deliveries- the warehouse manager needs to be able to update the stock.

Management and Marketing

And finally, management and the marketing team have their own portal to display metrics about stores and their sales.

This should include:

- Total sales
- Sales per category
- Best selling items
- Trending items (items with higher than usual sales)

These should be filterable, searchable and downloadable in a usable format.

Extensions and Improvements

In order to reach the highest marks for this project, you should complete not only the basic requirements but also one or more additional features which should be fully integrated into the basic system.

Note that full credit will not be given where these are simply implemented by using an existing package- these should involve some significant additional work.

You may suggest your own additional features to your product owner, or you could consider:

- Producing a native Android app in addition to the web app
- Integrating algorithmic product recommendations
- Graph generation for marketing